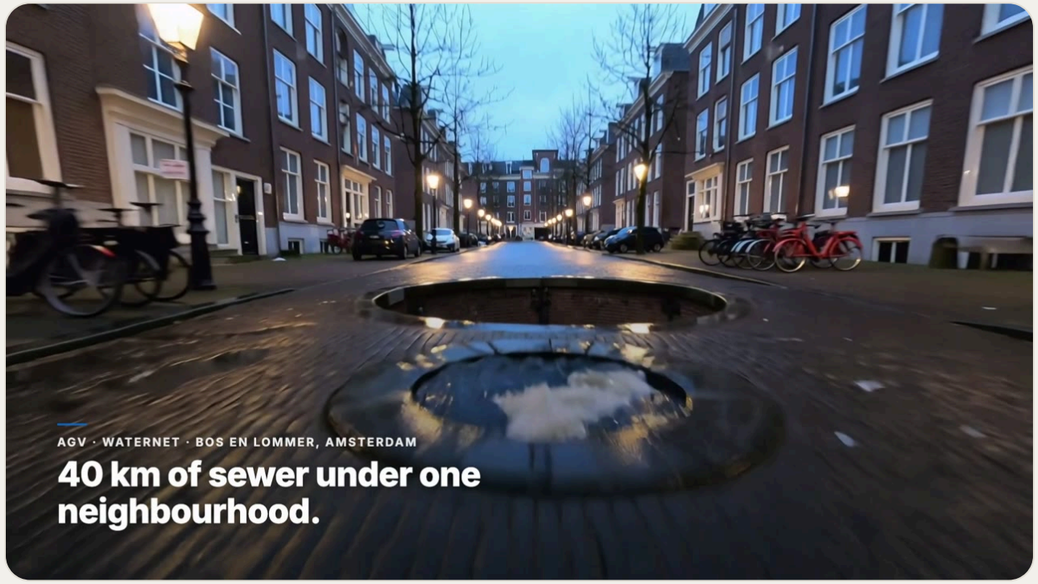
THE STORY · OPENING · 0:20

Under Amsterdam lie roughly 3,900 km of sewer. Whether to defer, reline or replace a pipe is decided in disconnected steps, without taking the climate into account.

Hack the Water Graph · Use Case 4 · Integrated intervention decision

THE IDEA IN ONE MINUTE



Watch the 60-second film

capncookdev.github.io/AGV-Hackathon · slide 2 of the online deck

THE STORY · THE CHALLENGE · 0:40

Waternet decides pipe by pipe, with the facts spread over systems that don’t communicate.

3

separate systems: cost in Maximo, risk in inspection data, function in GIS and IFC

Manual

work to bring them together into one view

No link

between financial, environmental and social costs

Use Case 4 brief, Waternet / AGV

THE STORY · OUR IDEA · 0:20

> **We help Waternet decide, for every sewer section, whether to defer, reline or replace, by putting financial, social and environmental factors in one euro value and letting a risk model advise the outcome.**

1

Knowledge Graph

One SPARQL query · 1,804 pipes

2

Three Values

Financial · Social · Environmental, in €

3

Risk Over Time

Condition 1–6 · Markov model · Soil

4

Advice + Why

Per pipe, in a browser and in IFC

THE PROOF · DEMO · 1:00

Our solution: for each pipe, an individual advice.

Every Pipe

Gets its own advice: defer, reline or replace.

One Price

Money, disruption and carbon, added up in euros for each option.

Explained

Click any pipe on the 3D map to see why.

What the demo shows

- Most pipes can wait: 1,210 of the 1,804 pipes in our demo area need nothing now.
- Where action is needed, the model mostly advises replacing (441) over relining (153).
- The demo covers part of Amsterdam. The same model is built to run on Waternet's whole network.

Open the live demo →

THE PROOF · UNDER THE HOOD · 1:00

What’s real, what’s assumed, what’s missing.

What’s real

- **Network:** W-OTL graph (SPARQL), NMD pipes & liners
- **Surroundings:** GeoTOP soil, BAG addresses, trees
- **Street works:** WIOR, Stadsregie, NDW
- **Environmental impact:** ecoinvent MKI

What’s inferred or synthetic

- **Condition and failure risk:** classes 1–6 simulated from age and material; failure rate per class is an assumed table
- **Unit costs:** published Barendrecht 2023 prices
- **Closure days and peat settlement:** estimated

What doesn’t work yet

- **Costs decide how, not whether:** switching cost source changes 11% of advice
- **Needed:** Maximo costs, inspection scores, failure history

Limitations in detail → · IFC 4.3 model with soil layers

THE PROOF · WHAT IT CHANGES · 0:40

Change the data and the advice moves. Environment is now part of the conversation.

Carbon costs more

Raise the CO_2 price and the advice shifts, pipe by pipe, with the reason shown for each.

The street opens anyway

When other works are planned there, digging and closure are shared, so replacing becomes cheaper.

A busy street

More homes nearby make each day of closure cost more, which favours the quicker reline.

For every pipe: what acting now instead of waiting a year is worth, in money, disruption and carbon.

THE CLOSE · 0:30

3,900 km of sewer. One question per pipe. Now answered with the climate in it.

Money, people and climate in one number, with the reason behind every defer, reline and replace. Add Waternet's own costs and inspections, and it runs on the whole network.

Open the live demo →

waterschap amstel, gooi en vecht

waternet
gemeente amsterdam

LINKS

Everything is online.

Pitch deck with film

capncookdev.github.io/AGV-Hackathon

Live demo with guided tour

.../dashboard/?tour=1

IFC 4.3 model of the network, with GeoTOP soil layers

.../dashboard/data/sewer.ifc

Limitations in detail

.../dashboard/slides/limitations.html

Source code, data pipeline and SPARQL query

github.com/CapncookDev/AGV-Hackathon

Hackathon prototype, built during Hack the Water Graph (Amsterdam, 13 November 2026). Not an official AGV or Waternet product.